Quadratic Minimization in Matrix Form: From Sums to Systems

A Bridge Document for SIMA — The Engine Behind OLS, Ridge, and Graduation

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1 Notation and Conventions

Throughout this document we use a consistent notation to distinguish scalars, vectors, and matrices. Refer to this table whenever a symbol is unclear.

Symbol	Meaning	Example
x, a, λ	Scalar (lowercase, plain)	$x \in \mathbb{R}$
$\boldsymbol{x}, \boldsymbol{a}, \boldsymbol{b}$	Column vector (lowercase, bold)	$\boldsymbol{x} \in \mathbb{R}^n$
$\boldsymbol{A}, \boldsymbol{X}, \boldsymbol{W}$	Matrix (uppercase, bold)	$\boldsymbol{A} \in \mathbb{R}^{n \times n}$
$f(x)$	Scalar-valued function	$f : \mathbb{R} \to \mathbb{R}$
$f(\boldsymbol{x})$	Scalar-valued function of a vector	$f : \mathbb{R}^n \to \mathbb{R}$
∇f	Gradient vector	$\nabla f \in \mathbb{R}^n$
$\boldsymbol{H}$	Hessian matrix	$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$
$\boldsymbol{A} \succ 0$	$\boldsymbol{A}$ is positive definite	$\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} > 0$ for all $\boldsymbol{x} \neq \boldsymbol{0}$
$\boldsymbol{A}'$ or $\boldsymbol{A}^\top$	Transpose of $\boldsymbol{A}$	rows become columns
$\boldsymbol{I}$	Identity matrix	$\boldsymbol{I}\boldsymbol{x} = \boldsymbol{x}$

Convention: All vectors are column vectors. A row vector is written $\boldsymbol{x}'$ (the transpose of a column). The product $\boldsymbol{a}'\boldsymbol{x}$ is a scalar (dot product). The product $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$ is also a scalar (quadratic form).

2 Scalar Warm-Up: Minimizing $f(x) = ax^2 + bx + c$

Before any matrices, let us solve the simplest optimization problem.

Scalar Quadratic Minimum

Given $f(x) = ax^2 + bx + c$ with $a > 0$:

$$f'(x) = 2ax + b = 0 \quad \Longrightarrow \quad x^* = -\frac{b}{2a}.$$

The second derivative is $f''(x) = 2a > 0$, confirming this is a minimum.

The Parabola Picture

The function $f(x) = ax^2 + bx + c$ is a parabola. When $a > 0$ the parabola opens upward, so it has a unique minimum at its vertex. Taking the derivative and setting it to zero locates that vertex. If $a < 0$ the parabola opens downward and the critical point is a *maximum* — there is no minimum.

The Template

This is the entire strategy. Every problem in this document follows the same three steps:

1. Write the objective as a quadratic function of the unknowns.
2. Take the derivative and set it to zero.
3. Solve the resulting equation for the unknowns.

The only thing that changes is the dimension: scalar, then 2D, then n-dimensional.

3 Two Variables: The Step to Linear Algebra

Consider a function of two variables:

$$f(x, y) = ax^2 + by^2 + cxy + dx + ey + f_0.$$

System of Two Equations

Setting both partial derivatives to zero:

$$\frac{\partial f}{\partial x} = 2ax + cy + d = 0, \tag{1}$$

$$\frac{\partial f}{\partial y} = 2by + cx + e = 0. \tag{2}$$

This is a system of two linear equations in two unknowns. We can write it in matrix form:

$$\underbrace{\begin{pmatrix} 2a & c \\ c & 2b \end{pmatrix}}_{\boldsymbol{A}} \underbrace{\begin{pmatrix} x \\ y \end{pmatrix}}_{\boldsymbol{x}} = \underbrace{\begin{pmatrix} -d \\ -e \end{pmatrix}}_{\boldsymbol{r}}$$

so the solution is $\boldsymbol{x}^* = \boldsymbol{A}^{-1}\boldsymbol{r}$, provided $\boldsymbol{A}$ is invertible.

A Bowl in 3D

The graph of $f(x, y)$ is a surface in three dimensions. When the quadratic part forms an elliptic paraboloid (a bowl), there is a unique minimum at the bottom. The two partial derivatives being zero means the surface is flat in both the x-direction and the y-direction simultaneously. The shape of the bowl (round vs. elongated) depends on the coefficients a, b, and c.

Notice how the two scalar equations (1)–(2) become a single matrix equation $\boldsymbol{Ax} = \boldsymbol{r}$. This is the key transition: *many equations become one matrix equation.*

4 The General Case: $f(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{Ax} + \boldsymbol{b}'\boldsymbol{x} + c$

Now we state the general result that covers all dimensions at once.

General Quadratic Minimization

Let $\boldsymbol{x} \in \mathbb{R}^n$, let $\boldsymbol{A}$ be an $n \times n$ symmetric matrix, let $\boldsymbol{b} \in \mathbb{R}^n$, and let c be a scalar. Define

$$f(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{Ax} + \boldsymbol{b}'\boldsymbol{x} + c.$$

The gradient is

$$\nabla f(\boldsymbol{x}) = 2\boldsymbol{Ax} + \boldsymbol{b}.$$

Setting the gradient to zero:

$$2\boldsymbol{Ax} + \boldsymbol{b} = \boldsymbol{0} \quad \Longrightarrow \quad \boldsymbol{x}^* = -\frac{1}{2}\boldsymbol{A}^{-1}\boldsymbol{b}.$$

Condition for a minimum: $\boldsymbol{A}$ must be positive definite ($\boldsymbol{A} \succ 0$).

Same Parabola, More Dimensions

This is exactly the scalar result $x^* = -b/(2a)$, but with bold letters. The matrix $\boldsymbol{A}$ plays the role of the scalar a, the vector $\boldsymbol{b}$ plays the role of the scalar b, and the inverse $\boldsymbol{A}^{-1}$ plays the role of dividing by a. Positive definite means "the bowl opens upward in every direction" — the n-dimensional analogue of $a > 0$.

Dimension Check

- $\boldsymbol{x}$ is $n \times 1$.
- $\boldsymbol{A}$ is $n \times n$, so $\boldsymbol{Ax}$ is $n \times 1$.
- $\boldsymbol{b}$ is $n \times 1$.
- The gradient $2\boldsymbol{Ax} + \boldsymbol{b}$ is $n \times 1$. Every term matches.
- $\boldsymbol{A}^{-1}$ is $n \times n$, so $\boldsymbol{A}^{-1}\boldsymbol{b}$ is $n \times 1$. The solution $\boldsymbol{x}^*$ is $n \times 1$. Consistent.

5 Matrix Calculus: The Key Rules

This section is the bridge that many students find difficult. We state just three rules; they are all we need.

Three Differentiation Rules

Let $\boldsymbol{x} \in \mathbb{R}^n$, $\boldsymbol{a} \in \mathbb{R}^n$ (constant), and $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ (constant, symmetric). Then:

Rule	Function	Gradient w.r.t. $\boldsymbol{x}$
1	$f(\boldsymbol{x}) = \boldsymbol{a}'\boldsymbol{x}$	$\nabla f = \boldsymbol{a}$
2	$f(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{Ax}$	$\nabla f = 2\boldsymbol{Ax}$
3	$f(\boldsymbol{x}) = \boldsymbol{x}'\boldsymbol{Ax} + \boldsymbol{b}'\boldsymbol{x}$	$\nabla f = 2\boldsymbol{Ax} + \boldsymbol{b}$

They Are the Same Rules You Already Know

Compare to the scalar case:

$$\frac{d}{dx}(ax) = a \quad \longleftrightarrow \quad \frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{a}'\boldsymbol{x}) = \boldsymbol{a} \qquad \text{(Rule 1)}$$

$$\frac{d}{dx}(ax^2) = 2ax \quad \longleftrightarrow \quad \frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}'\boldsymbol{Ax}) = 2\boldsymbol{Ax} \qquad \text{(Rule 2)}$$

$$\frac{d}{dx}(ax^2 + bx) = 2ax + b \quad \longleftrightarrow \quad \frac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}'\boldsymbol{Ax} + \boldsymbol{b}'\boldsymbol{x}) = 2\boldsymbol{Ax} + \boldsymbol{b} \qquad \text{(Rule 3)}$$

The structure is identical. Replace a with $\boldsymbol{A}$, x with $\boldsymbol{x}$, b with $\boldsymbol{b}$. The calculus works the same way — just with bold letters.

Verification of Rule 2 with a 2×2 Example

Let us verify Rule 2 by brute force. Take

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad \boldsymbol{A} = \begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}.$$

Expand the quadratic form:

$$\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} = a_{11}x_1^2 + 2a_{12}x_1x_2 + a_{22}x_2^2.$$

Differentiate component by component:

$$\frac{\partial}{\partial x_1}(\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}) = 2a_{11}x_1 + 2a_{12}x_2, \qquad \frac{\partial}{\partial x_2}(\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}) = 2a_{12}x_1 + 2a_{22}x_2.$$

In vector form:

$$\nabla(\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}) = 2\begin{pmatrix} a_{11} & a_{12} \\ a_{12} & a_{22} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = 2\boldsymbol{A}\boldsymbol{x}. \quad \checkmark$$

This confirms Rule 2. The matrix formula gives exactly the same result as differentiating term by term.

6 OLS as Quadratic Minimization

Ordinary Least Squares (OLS) regression is the most common application of this framework.

OLS Derivation

Data: observations $\boldsymbol{y} \in \mathbb{R}^n$, design matrix $\boldsymbol{X} \in \mathbb{R}^{n\times p}$.
Objective: minimize the sum of squared residuals:

$$S(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 = (\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta})'(\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}).$$

Expand:

$$S(\boldsymbol{\beta}) = \boldsymbol{y}'\boldsymbol{y} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\beta}'\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta}.$$

This is quadratic in $\boldsymbol{\beta}$ with:

$$\boldsymbol{A} = \boldsymbol{X}'\boldsymbol{X}, \qquad \boldsymbol{b} = -2\boldsymbol{X}'\boldsymbol{y}, \qquad c = \boldsymbol{y}'\boldsymbol{y}.$$

Gradient (using Rule 3):

$$\nabla_{\boldsymbol{\beta}} S = 2\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} - 2\boldsymbol{X}'\boldsymbol{y}.$$

Set to zero and solve:

$$2\boldsymbol{X}'\boldsymbol{X}\boldsymbol{\beta} - 2\boldsymbol{X}'\boldsymbol{y} = \boldsymbol{0} \quad \Longrightarrow \quad \boxed{\boldsymbol{\beta}^* = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}}$$

These are the **normal equations**.

The Bowl of Residuals

The function $S(\boldsymbol{\beta})$ is the total squared error as a function of the regression coefficients. Its graph is a bowl (paraboloid) in p-dimensional coefficient space. The OLS solution sits at the bottom of this bowl — the coefficient values that make the total squared error as small as possible.

Dimension Check

- $\boldsymbol{y}$: $n \times 1$ (one observation per row).
- $\boldsymbol{X}$: $n \times p$ (n observations, p predictors).
- $\boldsymbol{\beta}$: $p \times 1$ (one coefficient per predictor).
- $\boldsymbol{X}'\boldsymbol{X}$: $(p \times n)(n \times p) = p \times p$.

- $\boldsymbol{X}'\boldsymbol{y}$: $(p \times n)(n \times 1) = p \times 1$.
- $(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$: $(p \times p)(p \times 1) = p \times 1$. Matches $\boldsymbol{\beta}$.

7 Whittaker–Henderson as Quadratic Minimization

In actuarial science, Whittaker–Henderson graduation smooths observed mortality rates. The objective balances *fidelity to data* against *smoothness*.

Whittaker–Henderson Derivation

Data: observed rates $\boldsymbol{m} \in \mathbb{R}^n$, weight matrix $\boldsymbol{W} \in \mathbb{R}^{n \times n}$ (diagonal, positive), difference matrix $\boldsymbol{D} \in \mathbb{R}^{(n-z) \times n}$ (order z), smoothing parameter $\lambda > 0$.
Objective: minimize

$$S(\boldsymbol{g}) = \underbrace{(\boldsymbol{g} - \boldsymbol{m})'\boldsymbol{W}(\boldsymbol{g} - \boldsymbol{m})}_{\text{fidelity: stay close to data}} + \underbrace{\lambda\, \boldsymbol{g}'\boldsymbol{D}'\boldsymbol{D}\boldsymbol{g}}_{\text{smoothness: penalize roughness}}.$$

Expand:

$$\begin{aligned} S(\boldsymbol{g}) &= \boldsymbol{g}'\boldsymbol{W}\boldsymbol{g} - 2\boldsymbol{m}'\boldsymbol{W}\boldsymbol{g} + \boldsymbol{m}'\boldsymbol{W}\boldsymbol{m} + \lambda\, \boldsymbol{g}'\boldsymbol{D}'\boldsymbol{D}\boldsymbol{g} \\ &= \boldsymbol{g}'\big(\boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}\big)\boldsymbol{g} - 2\boldsymbol{m}'\boldsymbol{W}\boldsymbol{g} + \text{constant}. \end{aligned}$$

This is quadratic in $\boldsymbol{g}$ with:

$$\boldsymbol{A} = \boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}, \qquad \boldsymbol{b} = -2\boldsymbol{W}\boldsymbol{m}.$$

Gradient (using Rule 3):

$$\nabla_{\boldsymbol{g}} S = 2\big(\boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}\big)\boldsymbol{g} - 2\boldsymbol{W}\boldsymbol{m}.$$

Set to zero and solve:

$$\boxed{\boldsymbol{g}^* = \big(\boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}\big)^{-1}\boldsymbol{W}\boldsymbol{m}}$$

Same Bowl, Different Ingredients

The minimization machinery is identical to OLS. The only difference is *what* the matrix $\boldsymbol{A}$ contains:

- In OLS, $\boldsymbol{A} = \boldsymbol{X}'\boldsymbol{X}$ encodes the geometry of the data.
- In Whittaker–Henderson, $\boldsymbol{A} = \boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}$ combines two forces: the weight matrix $\boldsymbol{W}$ says "trust the data" and the penalty $\lambda \boldsymbol{D}'\boldsymbol{D}$ says "be smooth." The parameter λ controls the trade-off.

Both problems are bowls. Both solutions are found by taking a gradient and setting it to zero. The template is the same.

Dimension Check

- $\boldsymbol{g}$: $n \times 1$ (graduated rates, one per age).
- $\boldsymbol{m}$: $n \times 1$ (observed rates).
- $\boldsymbol{W}$: $n \times n$ (diagonal weight matrix).
- $\boldsymbol{D}$: $(n - z) \times n$ (difference operator of order z).
- $\boldsymbol{D}'\boldsymbol{D}$: $(n \times (n - z))((n - z) \times n) = n \times n$.

- $\boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}$: $n \times n$. Invertible (positive definite).
- $\boldsymbol{W}\boldsymbol{m}$: $(n \times n)(n \times 1) = n \times 1$.
- Solution: $(n \times n)^{-1}(n \times 1) = n \times 1$. Matches $\boldsymbol{g}$.

8 Ridge Regression as Quadratic Minimization

Ridge regression adds an L^2 penalty to OLS, preventing overfitting when predictors are correlated.

Ridge Derivation

Objective:

$$S(\boldsymbol{\beta}) = \|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2.$$

Expand:

$$S(\boldsymbol{\beta}) = \boldsymbol{\beta}'\big(\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I}\big)\boldsymbol{\beta} - 2\boldsymbol{y}'\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{y}'\boldsymbol{y}.$$

Solution:

$$\boxed{\boldsymbol{\beta}^* = \big(\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I}\big)^{-1}\boldsymbol{X}'\boldsymbol{y}}$$

The Unified View: Three Methods, One Template

Method	System matrix $\boldsymbol{A}$	Right-hand side	Solution
OLS	$\boldsymbol{X}'\boldsymbol{X}$	$\boldsymbol{X}'\boldsymbol{y}$	$(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$
Ridge	$\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I}$	$\boldsymbol{X}'\boldsymbol{y}$	$(\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{y}$
WH Graduation	$\boldsymbol{W} + \lambda\boldsymbol{D}'\boldsymbol{D}$	$\boldsymbol{W}\boldsymbol{m}$	$(\boldsymbol{W} + \lambda\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{W}\boldsymbol{m}$

All three are instances of the same template: given a quadratic objective in some vector of unknowns, take the gradient, set it to zero, and solve the resulting linear system $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{r}$. The mathematics does not care whether $\boldsymbol{x}$ represents regression coefficients or graduated mortality rates.

9 Positive Definiteness: Why It Guarantees a Minimum

Positive Definiteness

A symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ is **positive definite** if

$$\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} > 0 \quad \text{for all } \boldsymbol{x} \neq \boldsymbol{0}.$$

Equivalent characterizations:

- All eigenvalues of $\boldsymbol{A}$ are strictly positive.
- $\boldsymbol{A}$ is invertible (so the solution $\boldsymbol{A}^{-1}\boldsymbol{r}$ exists and is unique).
- The quadratic form $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}$ is strictly convex.

Key facts for our applications:

- $\boldsymbol{X}'\boldsymbol{X}$ is always positive *semi*-definite. It is positive definite when $\boldsymbol{X}$ has full column rank (more observations than predictors, no perfect collinearity).

- $\boldsymbol{X}'\boldsymbol{X} + \lambda \boldsymbol{I}$ is always positive definite for $\lambda > 0$, even when $\boldsymbol{X}'\boldsymbol{X}$ is singular. This is why Ridge regression "fixes" collinearity.
- $\boldsymbol{W} + \lambda \boldsymbol{D}'\boldsymbol{D}$ is positive definite when $\boldsymbol{W}$ has enough positive diagonal entries, which is always the case in practice (we observe data at those ages).

The Bowl Must Open Upward

Positive definite means the bowl opens upward *in every direction*. There is exactly one bottom point, and the gradient equation finds it.
If $\boldsymbol{A}$ were *not* positive definite, the surface could have:
- A saddle point (opens up in some directions, down in others) — no minimum.
- A flat valley (zero eigenvalue) — infinitely many minima, no unique solution.

Positive definiteness rules out both pathologies and guarantees a unique minimum.

10 Summary Table

Problem	Objective	$\boldsymbol{A}$ matrix	RHS	Solution
Scalar	$ax^2 + bx$	a	$-b/2$	$-b/(2a)$
OLS	$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2$	$\boldsymbol{X}'\boldsymbol{X}$	$\boldsymbol{X}'\boldsymbol{y}$	$(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$
Ridge	$\|\boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|^2$	$\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I}$	$\boldsymbol{X}'\boldsymbol{y}$	$(\boldsymbol{X}'\boldsymbol{X} + \lambda\boldsymbol{I})^{-1}\boldsymbol{X}'\boldsymbol{y}$
WH Grad.	$(\boldsymbol{g} - \boldsymbol{m})'\boldsymbol{W}(\boldsymbol{g} - \boldsymbol{m}) + \lambda\boldsymbol{g}'\boldsymbol{D}'\boldsymbol{D}\boldsymbol{g}$	$\boldsymbol{W} + \lambda\boldsymbol{D}'\boldsymbol{D}$	$\boldsymbol{W}\boldsymbol{m}$	$(\boldsymbol{W} + \lambda\boldsymbol{D}'\boldsymbol{D})^{-1}\boldsymbol{W}\boldsymbol{m}$

Every row is the same algorithm: identify the quadratic structure, differentiate, set to zero, solve. The column headings are the only things that change.

11 Key Identities

For quick reference, the identities used throughout this document:

(i) $\dfrac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{a}'\boldsymbol{x}) = \boldsymbol{a}$ (gradient of linear form)

(ii) $\dfrac{\partial}{\partial \boldsymbol{x}}(\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x}) = 2\boldsymbol{A}\boldsymbol{x}$ ($\boldsymbol{A}$ symmetric; gradient of quadratic form)

(iii) $(\boldsymbol{A}\boldsymbol{B})' = \boldsymbol{B}'\boldsymbol{A}'$ (transpose of a product reverses order)

(iv) $\boldsymbol{A} \succ 0 \implies \boldsymbol{A}$ is invertible (positive definite implies invertible)

(v) The quadratic $\boldsymbol{x}'\boldsymbol{A}\boldsymbol{x} + \boldsymbol{b}'\boldsymbol{x} + c$ with $\boldsymbol{A} \succ 0$ has its unique minimum at

$$\boldsymbol{x}^* = -\tfrac{1}{2}\boldsymbol{A}^{-1}\boldsymbol{b}.$$

12 What This Document Does Not Cover

This document is deliberately limited to *unconstrained quadratic minimization with a closed-form solution*. The following related topics are important but outside our scope:

- **Non-quadratic optimization.** When the objective is not quadratic (e.g., maximum likelihood with non-normal distributions), there is no closed-form solution. Iterative methods such as Newton's method or gradient descent are needed.

- **Constrained optimization.** When the unknowns must satisfy constraints (e.g., probabilities summing to 1, non-negative rates), the method of Lagrange multipliers or KKT conditions applies. The gradient-equals-zero condition must be modified.
- **Numerical methods for solving the linear system.** We write $\boldsymbol{x}^* = \boldsymbol{A}^{-1}\boldsymbol{r}$, but in practice one does not compute $\boldsymbol{A}^{-1}$ explicitly. Efficient algorithms include Cholesky decomposition (exploiting symmetry and positive definiteness), LU factorization, and banded solvers (exploiting the sparsity of $\boldsymbol{D}'\boldsymbol{D}$ in Whittaker–Henderson).